Facade

Season: 2 · Structural · Category: GoF · Level: Beginner

Intent

- Provide a simple interface to a noisy subsystem.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Hotel concierge books taxi, dinner, tickets.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- When a simpler redesign removes the need.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```
package com.buildwithwaqar.patterns.gof.structural.facade;

final class Inventory { boolean reserve(String sku, int qty) { return qty > 0 && !sku.isBlank(); } }
final class PaymentGateway { boolean charge(String card, int cents) { return card != null && cents > 0; } }
final class Shipping { String ship(String sku) { return "TRACK-" + sku; } }

/** Simple interface over a noisy subsystem. */
public final class OrderFacade {
    private final Inventory inventory = new Inventory();
    private final PaymentGateway payments = new PaymentGateway();
    private final Shipping shipping = new Shipping();

    public String placeOrder(String sku, int qty, String card, int cents) {
        if (!inventory.reserve(sku, qty)) return "REJECTED:inventory";
        if (!payments.charge(card, cents)) return "REJECTED:payment";
        return "OK:" + shipping.ship(sku);
    }
}
```

Modern Java 25 / Functional

- Prefer sealed interfaces + records; virtual proxy uses synchronized lazy init.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/structural/facade/
```

Pitfalls

- Keep adapters thin.
- Decorators should preserve the interface.
- Facades must not become god objects.

Interview Q&A

- Q: When is Facade the wrong choice? A: When you can change the adaptee/subject API or collapse the hierarchy.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Provide a simple interface to a noisy subsystem.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Hotel concierge books taxi, dinner, tickets. [0:40] Intent — State the design force in one sentence:...